TEAM 9 • PES UNIVERSITY

SOFTWARE TEST PLAN

Electronics Store Inventory Management System

Updated verification strategy, test cases and requirements traceability

Document control	Details
Document ID	EIMS-STP-1.1
Version	1.1 — Sprint 2 enhancement revision
Date	03-11-2026
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Status	Updated draft for review and evaluator sign-off
Repository	ElectronicsStore-ManagementSystem / Team-9-Electronics-Store-Inventory-Management-System

Revision purpose. Preserve the September 2026 baseline while extending the requirements, design and verification coverage to include functionality added during subsequent development sprints.

Revision history

Version	Date	Change summary	Approval
1.0	16-09-2026	Original test plan for baseline SRS requirements.	Pending in source document
1.1	03-11-2026	Extends scope, test items, cases, RTM and regression coverage for later implementation features through F-032.	For QA/team/guide review

1. Introduction

1.1 Purpose. Define the test approach, test environment, entry/exit criteria, test cases, security validation, traceability, responsibilities, risks and deliverables for EIMS v1.1.

1.2 Scope. Test original inventory, component creation/deletion, reports, dashboard, authentication/RBAC, audit logging and NFR/security requirements, plus supplier and branch management, image lifecycle, stock movement history, purchase orders, system availability reporting, persistent theme and structured API errors.

1.3 References. EIMS SRS v1.1; SAD v1.1; repository docs/API.md and docs/ARCHITECTURE.md; implementation and test code in the project repository.

1.4 Test levels. Unit, API/integration, security, UI/manual smoke, regression and performance/scalability testing. A test run must record build/commit, environment, tester, date, test data, actual result and evidence.

2. Test items

Inventory list/search/filter/sort/update; component creation, SKU generation and validation; deletion and audit; supplier catalog and branch management; component image upload/preview/replace/remove; stock-in/stock-out and movement history; purchase-order creation/status/receiving/receipt history; availability reports and exports; system availability calculations; dashboard; authentication, session expiry and RBAC; responsive layout and persistent theme; API error handling.

3. Test approach and strategy

3.1 Functional testing. Verify positive, negative, boundary and authorization paths for each functional requirement. Verify persisted state, not only frontend messages.

3.2 Integration testing. Verify API-to-service-to-database workflows, especially atomic stock changes, purchase-order receiving, movement history and report consistency.

3.3 Security testing. Verify authentication, role enforcement, password hashing, session expiry, transport configuration and injection/XSS protections. Run security tests using isolated non-production data.

3.4 Regression testing. Re-run impacted test suites after changes to inventory, suppliers, branches, movements, POs, reports, authentication or shared frontend behaviour.

3.5 Performance and usability. Use the SRS thresholds for dashboard/inventory response, search, 10,000 components, 20 concurrent users and responsive viewport checks. Record tool, hardware, dataset and results.

3.6 Test design techniques. Equivalence partitioning, boundary-value analysis, decision tables for role/status combinations, state-transition testing for purchase-order status, and negative testing for malformed requests.

4. Test environment, data and criteria

4.1 Environment. Python 3.12 as configured by CI; FastAPI application; SQLAlchemy/SQLite local test database; pytest; browser-based UI checks; Node syntax/security tests where applicable. Production HTTPS must be checked in the actual deployment environment.

4.2 Test data. Seeded Admin/Manager and Staff accounts; active/inactive components; valid and invalid SKUs; multiple categories, suppliers and branches; low/out-of-stock items; purchase orders with multiple line items; partial/full receipt examples; image files of supported and unsupported types and sizes; monitoring records including scheduled maintenance and missing-data scenarios.

4.3 Entry criteria. Stable build available; migrations/bootstrap complete; test accounts and test data available; API and database reachable; test tools installed; scope and acceptance criteria reviewed.

4.4 Exit criteria. All planned tests executed; no unresolved critical defect affecting core inventory, authentication, authorization, stock integrity or purchase-order receiving; failures triaged; RTM updated; NFR results recorded; deviations accepted by the team/guide.

5. Test cases — original and added functionality

Test ID	Req ID	Test area	Procedure / scenario	Expected result	Level
TC-INV-01	F-001	Inventory list	Verify active components and required attributes are displayed.	Expected: active records and fields match database.	Integration/UI
TC-INV-02	F-002	Search	Search by name, SKU and category; include no-match case.	Expected: correct results; record response time against 2-second target.	Integration
TC-INV-03	F-003	Filters	Apply category, supplier and stock-status filters individually and together.	Expected: only matching records are returned.	Integration/UI
TC-INV-04	F-004	Sorting	Sort quantity, price and date-added ascending/descending.	Expected: correct ordering.	UI/API
TC-INV-05	F-005	Update component	Update quantity, price, supplier and reorder level.	Expected: persisted values are reflected in list/detail.	Integration
TC-ADD-01	F-006	Add component	Submit valid component data.	Expected: component is created and visible.	Integration
TC-ADD-02	F-007	SKU generation	Create multiple components and inspect generated identifiers.	Expected: unique identifiers; deleted SKU is not reused if this remains the implemented rule.	Integration
TC-ADD-03	F-008	Required-field validation	Submit missing, blank, negative and malformed values.	Expected: rejected with clear field-level errors.	UI/API
TC-ADD-04	F-009	Duplicate SKU	Attempt duplicate SKU insertion.	Expected: duplicate rejected; existing record unchanged.	Integration
TC-ADD-05	F-010	Image upload	Upload valid image and verify association/display.	Expected: accepted image displayed for component.	Integration/UI
TC-DEL-01	F-011	Delete component	Delete an eligible component as authorized role.	Expected: no longer appears in active inventory.	Integration
TC-DEL-02	F-012	Delete confirmation	Cancel and confirm deletion.	Expected: cancel preserves record; confirm deletes.	UI
TC-DEL-03	F-013	Delete authorization	Attempt restricted deletion as Staff.	Expected: forbidden response; data unchanged.	Security/API
TC-DEL-04	F-014	Deletion audit	Delete component and inspect audit record.	Expected: user, timestamp and component recorded.	Integration
TC-REP-01	F-015	Availability classification	Create in-stock, low-stock and out-of-stock examples.	Expected: classifications follow reorder thresholds.	Integration
TC-REP-02	F-016	Report filters	Filter report by supported category/supplier/date criteria.	Expected: report contains matching rows only.	Integration/UI
TC-REP-03	F-017	Report exports	Export supported PDF, Excel and CSV formats.	Expected: files open and data matches report.	Integration/UI
TC-REP-04	F-018	Low-stock highlighting	Create item below reorder level.	Expected: item is highlighted in report/UI.	UI
TC-DASH-01	F-019–022	Dashboard	Compare totals, stock value, recent items, charts and low-stock alerts to database.	Expected: values and alerts are consistent.	Integration/UI
TC-SUP-01	F-023	Supplier catalog	Create supplier with valid data; retrieve catalog; attempt invalid data.	Expected: persisted supplier is selectable; invalid input rejected.	Integration
TC-BR-01	F-024	Branch management	Create/list/update branch; test invalid input and unauthorized action.	Expected: authorized changes persist; unauthorized action rejected.	Integration/Security
TC-IMG-01	F-025	Image lifecycle	Preview, replace and remove image; upload unsupported type and file over 5 MB.	Expected: lifecycle actions work; invalid files rejected.	Integration/UI
TC-UI-01	F-026	Add form UX	Inspect required indicators, inline errors, cancel/reset and responsive layout.	Expected: errors are clear; cancel resets unsaved values.	UI

Test ID	Req ID	Test area	Procedure / scenario	Expected result	Level
TC-FLT-01	F-027	Dynamic filters	Add supplier/category and load filter choices; combine filters.	Expected: choices reflect persisted records; result set is correct.	Integration/UI
TC-MOV-01	F-028	Stock in	Record valid stock-in quantity with reason.	Expected: quantity increases and movement metadata is stored.	Integration
TC-MOV-02	F-028	Stock out	Record valid stock-out quantity with reason.	Expected: quantity decreases and movement metadata is stored.	Integration
TC-MOV-03	F-028	Invalid/insufficient stock	Submit zero/negative quantity and stock-out greater than available quantity.	Expected: rejected; quantity and history unchanged.	Integration
TC-MOV-04	F-028	Movement history/RBAC	Retrieve component history as Staff/Manager; attempt unauthorized write or modification.	Expected: authorized read/write policy enforced; history cannot be altered by unauthorized user.	Security/Integration
TC-MOV-05	F-028	Movement rollback	Force failure during stock operation.	Expected: no partial quantity update or orphan movement record.	Integration
TC-PO-01	F-029	Create purchase order	Create PO for existing supplier with multiple component lines.	Expected: unique PO number and correct line items.	Integration
TC-PO-02	F-029	Status transitions	Change PO status through permitted transitions; attempt invalid transition.	Expected: valid transitions persist; invalid transition rejected.	Integration
TC-PO-03	F-029	Partial receiving	Receive part of ordered quantity.	Expected: received quantity and inventory update accurately; PO remains partially received as applicable.	Integration
TC-PO-04	F-029	Full receiving	Receive remaining quantities.	Expected: inventory updated and PO reaches completed/received state as designed.	Integration
TC-PO-05	F-029	Idempotency/rollback	Repeat same receive request and simulate failed receipt.	Expected: no duplicate stock increment; failed transaction rolls back.	Integration
TC-PO-06	F-029	Receipt history/RBAC	View receipts and attempt restricted PO operations.	Expected: receipt history is accurate and authorization enforced.	Security/Integration
TC-AV-01	F-030/NF-002	Availability data gaps	Test no monitoring records, incomplete coverage and scheduled maintenance.	Expected: controlled result and explicit coverage treatment.	Integration
TC-AV-02	F-030	Unexpected reporting error	Trigger unexpected reporting failure.	Expected: stable error response; no stack trace or sensitive internals.	API
TC-THEME-01	F-031	Theme persistence	Switch light/dark mode and reload page.	Expected: chosen theme persists and controls remain legible.	UI
TC-ERR-01	F-032	API error contract	Test validation, not-found, unauthorized and unexpected error responses.	Expected: appropriate status and consistent safe response shape.	API/Security
TC-SEC-01	SR-001	Authentication	Call protected API without session and with valid credentials.	Expected: unauthorized rejected; valid login accepted.	Security
TC-SEC-02	SR-002	Password hashing	Inspect stored password representation and verify login.	Expected: salted one-way hash; no plaintext password.	Security
TC-SEC-03	SR-003	RBAC	Exercise protected actions using Admin/Manager and Staff accounts.	Expected: permissions enforced server-side.	Security
TC-SEC-04	SR-004	Transport security	Inspect deployed endpoint and TLS configuration.	Expected: HTTPS/TLS 1.2+ in production.	Deployment review
TC-SEC-05	SR-005	Session expiry	Remain inactive beyond 15 minutes and call protected endpoint.	Expected: session rejected/expired.	Security
TC-SEC-06	SR-006	Injection/XSS	Submit SQL injection strings, malicious sort keys and HTML/script payloads.	Expected: inputs rejected/handled safely; output encoded; no execution.	Security
TC-NF-01	NF-001	Page performance	Measure dashboard/inventory page requests under normal load.	Expected: at least 90% within 3 seconds.	Performance

Test ID	Req ID	Test area	Procedure / scenario	Expected result	Level
TC-NF-02	NF-002	Monthly availability	Verify monthly calculation with monitored periods and scheduled maintenance.	Expected: result matches defined calculation and coverage assumptions.	Analysis
TC-NF-03	NF-003	Usability	Observe a new staff user completing add/search task without formal training.	Expected: task completed within five minutes.	Usability
TC-NF-04	NF-004	Scale/concurrency	Run 10,000-component dataset and 20 concurrent users.	Expected: results recorded and target met without unacceptable degradation.	Performance
TC-NF-05	NF-005	Maintainability review	Review module boundaries, docs and testability.	Expected: UI/API/data access separation and documentation present.	Review
TC-NF-06	NF-006	Responsive UI	Check desktop >=1280 px and tablet >=768 px.	Expected: no critical overflow or inaccessible controls.	UI

5.1 Security validation

Security testing shall cover all original security requirements and newly added endpoints. In addition to TC-SEC-01 through TC-SEC-06, verify that unauthorized users cannot create/receive purchase orders, create stock movements, manage branches/suppliers or alter historical movement/receipt records. Validate file upload type/size handling and ensure error responses do not disclose stack traces, secrets or database details.

Use non-production accounts and isolated test data. Record request/response evidence with credentials and tokens redacted. A passing local test does not prove deployed HTTPS; record deployment evidence separately.

6. Test schedule and responsibilities

The original September schedule is superseded for the enhancement regression cycle. The project record indicates Sprint 1 ran 19–25 October 2026 and Sprint 2 ran 26 October–1 November 2026. The exact dates for final regression, evaluator demonstration and sign-off must be entered by the team.

Role	Assigned person	Responsibility
QA Lead	Hemavathi E	Maintain RTM, coordinate test execution, triage defects and prepare test summary.
Test Engineer	Jayashree D	Prepare data, execute assigned cases, capture actual results/evidence and log defects.
Developer	Krishitha Kankanala	Support environment, investigate defects, implement fixes and participate in regression.
Product Owner / Team Lead	Karan Maheshwari	Confirm scope, prioritize defects, coordinate acceptance and sign-off.
All team members	Team 9	Demonstrate assigned implementation and explain related tests/PRs.

7. Requirements Traceability Matrix (RTM)

The table below maps each requirement group to its test coverage. The test IDs in Section 5 provide the detailed procedures. Mark execution status only after a test run is recorded.

Requirement	Coverage	Test case(s)	Execution status
F-001–F-005	Inventory list/search/filter/sort/update	TC-INV-01 to TC-INV-05	Not yet recorded
F-006–F-010	Add component/SKU/validation/image upload	TC-ADD-01 to TC-ADD-05	Not yet recorded
F-011–F-014	Deletion/confirmation/authorization/audit	TC-DEL-01 to TC-DEL-04	Not yet recorded
F-015–F-018	Availability report/filter/export/low-stock	TC-REP-01 to TC-REP-04	Not yet recorded
F-019–F-022	Dashboard widgets/charts/alerts	TC-DASH-01	Not yet recorded
F-023	Supplier catalog	TC-SUP-01	Not yet recorded
F-024	Branch management	TC-BR-01	Not yet recorded

Requirement	Coverage	Test case(s)	Execution status
F-025	Image preview/replace/remove	TC-IMG-01	Not yet recorded
F-026	Add form usability	TC-UI-01	Not yet recorded
F-027	Dynamic supplier/category filters	TC-FLT-01	Not yet recorded
F-028	Stock movement history	TC-MOV-01 to TC-MOV-05	Not yet recorded
F-029	Purchase order management	TC-PO-01 to TC-PO-06	Not yet recorded
F-030 / NF-002	System availability/report error handling	TC-AV-01 to TC-AV-02; TC-NF-02	Not yet recorded
F-031	Persistent theme	TC-THEME-01	Not yet recorded
F-032	Structured API errors	TC-ERR-01	Not yet recorded
SR-001–SR-006	Security controls	TC-SEC-01 to TC-SEC-06	Not yet recorded
NF-001–NF-006	Quality attributes	TC-NF-01 to TC-NF-06	Not yet recorded

8. Test deliverables and reporting

Deliverables: approved Test Plan; test data; automated test suite; manual test execution log; defect log; RTM with pass/fail/blocked status; performance and scalability measurements; security validation evidence; UI screenshots/smoke-test evidence; and final Test Summary Report.

Repository PR descriptions report historical suite counts at different commits, including 180 and 181 passing tests in later PRs. These are commit-specific claims, not a substitute for a fresh final run against the frozen release candidate. Record the exact final commit SHA and actual test output before sign-off.

9. Defect management, risks and suspension/resumption

Defects shall include ID, severity, affected requirement/test, environment, reproduction steps, expected/actual result, evidence, owner and resolution commit. Critical defects in authentication, authorization, inventory quantity integrity or duplicate PO receiving block release unless formally accepted by the guide.

Risks include unstable build, unavailable API/database, incorrect role configuration, inconsistent shared test data, missing monitoring data, and incomplete performance environment. Suspend testing when a blocking defect or environment failure invalidates results; resume after fix, data restoration and smoke test.

10. Test execution record template

For each test record: Test ID; requirement ID; build/commit SHA; tester; execution date/time; environment; preconditions; test data; steps; expected result; actual result; status (Pass/Fail/Blocked/Not Run); defect ID; evidence link; retest result. The current document defines planned coverage; it does not assert that every listed case has already been executed.